Inner Product Spaces, $\vec{\S}$6.7

Recall from previously:

- **Gram-Schmidt Orthogonalization.** Given a basis $\{x_1, \ldots, x_p\}$ for a non-zero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned}
v_1 &= x_1 \\
v_2 &= x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1}\right) v_1 \\
v_3 &= x_3 - \left(\frac{x_3 \cdot v_1}{v_1 \cdot v_1}\right) v_1 - \left(\frac{x_3 \cdot v_2}{v_2 \cdot v_2}\right) v_2 \\
&\vdots \\
v_p &= x_p - \left(\frac{x_p \cdot v_1}{v_1 \cdot v_1}\right) v_1 - \cdots - \left(\frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}}\right) v_{p-1}.
\end{aligned}$$

  Then $\{v_1, \ldots, v_p\}$ is an orthogonal basis for W. In addition $\text{Span}\{v_1, \ldots, v_k\} = \text{Span}\{x_1, \ldots, x_k\}$ for all $1 \leq k \leq p$.

- **QR Factorization.** If A is an $m \times n$ matrix with linearly independent columns, then $A = QR$ where Q is an $m \times n$ matrix whose columns form an orthonormal basis for the column space of A and R is an upper triangular $n \times n$ matrix with positive entries along the diagonal.

- A **least squares solution** to $Ax = b$ is a vector $\hat{x} \in \mathbb{R}^n$ such that

$$\|b - A\hat{x}\| \leq \|b - Ax\|$$

  for all $x \in \mathbb{R}^n$.

- **Theorem 6.13:** The set of least squares solutions to $Ax = b$ is the same as the set of solutions to the normal squares $A^T Ax = A^T b$.

- The **least squares error** is given by the distance from b to $A\hat{x}$.

Definition 1. *An **Inner Product Space** (over $\mathbb{R}$) is a vector space V with a pairing (called **the inner product**) $\langle \cdot, \cdot \rangle : V^2 \to \mathbb{R}$, that satisfies the following:*

- ***Linearity in the first factor:***
  $\langle c_1 f_1 + c_2 f_2, g \rangle = c_1 \langle f_1, g \rangle + c_2 \langle f_2, g \rangle$.
- ***Symmetry:*** $\langle g, f \rangle = \langle f, g \rangle$.
- ***Positive definiteness:*** $\langle f, f \rangle \geq 0$, *with* $\langle f, f \rangle = 0$ *if and only if* $f = 0$.

1. A few examples of inner product spaces.

(a) The set $\mathbb{R}^n$ with the inner product $\langle \vec{x}, \vec{y} \rangle = \vec{x} \cdot \vec{y}$.

(b) The set of polynomials of degree 4 or less, $\mathbb{P}_4$ with

$$\langle p, q \rangle = p(-2)q(-2) + \cdots + p(2)q(2).$$

(c) The space ℓ^2 of infinite sequences of real numbers $\vec{x} = (x_1, x_2, \ldots, x_n, \ldots)$ for which $\sum x_j^2$ converges, with inner product,

$$\langle \vec{x}, \vec{y} \rangle = \sum_{j=1}^{\infty} x_j y_j.$$

(d) The space $\mathrm{C}[a, b]$ of real valued continuous functions with domain $[a, b]$. The inner product is:

$$\langle f, g \rangle = \int_a^b f(x)g(x)dx.$$

2. Let $V = C[-1, 1]$ with inner product defined above. Let $W = \text{Span}\{1, t, t^2\}$.

(a) Find an orthogonal basis for W.

(b) Find the best approximation to e^t in W.

Consider $f \in C[0, 2\pi]$, then we want to approximate f using sinusoidal functions. The set

$$W_n = \{1, \cos t, \ldots, \cos nt, \sin t, \ldots, \sin nt\}$$

is orthogonal with respect to

$$\langle f, g \rangle = \int_0^{2\pi} f(t)g(t)dt.$$

Definition 2. *The n-th order* **Fourier approximation** *to $f \in C[0, 2\pi]$ is given by the $proj_{W_n} f$.*

Note: For $k > 0$, the **Fourier coefficients** a_k and b_k are given by

$$a_k = \frac{\langle f, \cos kt \rangle}{\langle \cos kt, \cos kt \rangle}, \text{ and } b_k = \frac{\langle f, \sin kt \rangle}{\langle \sin kt, \sin kt \rangle},$$

and hence

$$a_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos ktdt, \text{ and } b_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin ktdt.$$

3. Find the n-th order Fourier approximation for $f(t) = t$ on $[0, 2\pi]$.

Remark: More generally, the **Fourier series** is

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos mt + b_m \sin mt).$$